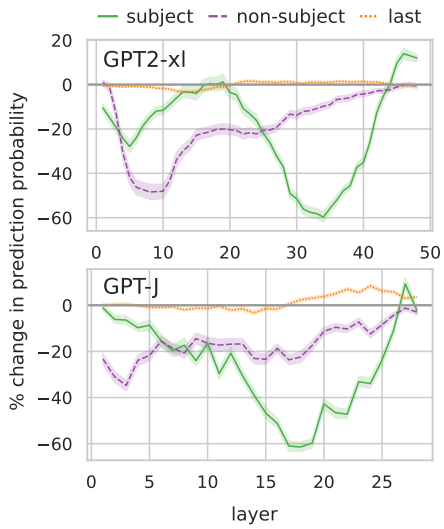

Figure 2: Relative change in the prediction probability when intervening on attention edges to the last position, for window sizes of 9 layers in GPT-2 and 5 in GPT-J.

5 Localizing Information Flow via Attention Knockout

For a successful attribute prediction, a model should process the input subject and relation such that the attribute can be read from the last position. We investigate how this process is done internally by "knocking out" parts of the computation and measuring the effect on the prediction. To this end, we propose a fine-grained intervention on the MHSA sublayers, as they are the only module that communicates information between positions, and thus any critical information must be transferred by them. We show that factual predictions are built in stages where critical information propagates to the prediction at specific layers during inference.

Method: Attention Knockout Intuitively, critical attention edges are those that, when blocked, result in severe degradation in prediction quality. Therefore, we test whether critical information propagates between two hidden representations at a specific layer, by zeroing-out all the attention edges between them. Formally, let $r, c \in [1, N]$ such that $r \leq c$ be two positions. We block $\mathbf{x}_r^\ell$ from attending to $\mathbf{x}_c^\ell$ at a layer $\ell < L$ by updating the attention weights to that layer (Eq. 5):

$$M_{rc}^{\ell+1,j} = -\infty \quad \forall j \in [1, H] \tag{7}$$

Effectively, this restricts the source position from obtaining information from the target position, at that particular layer. Notably, this is different from causal tracing (Meng et al., 2022a), which checks what hidden representations restore the original prediction when given perturbed input tokens; we test where critical information *propagates* rather than where it is *located* during inference.

Experiment We use Attention Knockout to test whether and, if so, where information from the subject and relation positions directly propagates to the last position. Let $\mathcal{S}, \mathcal{R} \subset [1, N)$ be the subject and non-subject positions for a given input. For each layer ℓ, we block the attention edges from the last position to each of $\mathcal{S}$, $\mathcal{R}$ and the last (N-th) position, for a window of k layers around the ℓ-th layer, and measure the change in prediction probability. We set $k = 9$ (5) for GPT-2 (GPT-J).[2]

Results Fig. 2 shows the results. For both GPT-2 and GPT-J, blocking attention to the subject tokens (solid green lines) in the middle-upper layers causes a dramatic decrease in the prediction probability of up to 60%. This suggests that critical information from the subject positions moves directly to the last position at these layers. Moreover, another substantial decrease of 35%-45% is observed for the non-subject positions (dashed purple lines). Importantly, critical information from non-subject positions precedes the propagation of critical information from the subject positions, a trend we observe for different subject-relation orders (§A.1). Example interventions are provided in §H.

Overall, this shows that there are specific disjointed stages in the computation with peaks of critical information propagating directly to the prediction from different positions. In the next section, we investigate the critical information that propagates from the subject positions to the prediction.

6 Intermediate Subject Representations

We saw that critical subject information is passed to the last position in the upper layers. We now analyze what information is contained in the subject representation at the point of transfer, and how does this information evolves across layers. To do this, we map hidden representations to vocabulary tokens via projection. Our results indicate that the subject representation contains a wealth of information about the subject at the point where it is transferred to the last position.

[2]Results for varying values of k are provided in §A.4.

	Subject	Example top-scoring tokens by the subject representation
GPT-2	Iron Man	`3, 2, Marvel, Ultron, Avenger, comics, suit, armor, Tony, Mark, Stark, 2020`
	Sukarno	`Indonesia, Buddhist, Thailand, government, Museum, Palace, Bangkok, Jakarta`
	Roman Republic	`Rome, Augustus, circa, conquered, fame, Antiqu, Greece, Athens, AD, Caesar`
GPT-J	Ferruccio Busoni	`music, wrote, piano, composition, International, plays, manuscript, violin`
	Joe Montana	`career, National, football, NFL, Award, retired, quarterback, throws, Field`
	Chromecast	`device, Audio, video, Wireless, HDMI, USB, Google, Android, technology, 2016`

Table 1: Example tokens by subject representations of GPT-2 ($\ell = 40$) and GPT-J ($\ell = 22$).

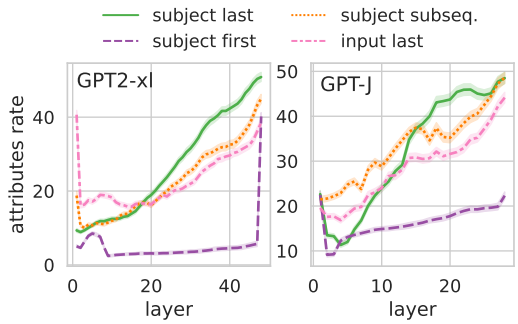

Figure 3: Attributes rate at different positions across layers (starting from layer 1), in GPT-2 and GPT-J.

6.1 Inspection of Subject Representations

Motivating Observation To analyze what is encoded in a representation $\mathbf{h}_t^\ell$, we cast it as a distribution $\mathbf{p}_t^\ell$ over the vocabulary (Geva et al., 2022b; Nostalgebraist, 2020), using the same projection applied to final-layer representations (Eq. 2). Then, we inspect the k tokens with the highest probabilities in $\mathbf{p}_t^\ell$. Examining these projections, we observed that they are informative and often encode several subject attributes (Tab. 1 and §D). Therefore, we turn to quantitatively evaluate the extent to which the representation of a subject encodes tokens that are semantically related to it.

Evaluation Metric: Attributes Rate Semantic relatedness is hard to measure based on human judgment, as ratings are typically of low agreement, especially between words of various parts of speech (Zesch and Gurevych, 2010; Feng et al., 2017). Hence, we propose an automatic approximation of the subject-attribute relatedness, which is the rate of the predicted attributes in a given set of tokens known to be highly related to the subject. For a given subject s, we first create a set $\mathcal{A}_s$ of candidate attributes, by retrieving paragraphs about s from Wikipedia using BM25 (Robertson et al., 1995), tokenizing each paragraph, and removing common words and sub-words. The set $\mathcal{A}_s$ consists of non-common tokens that were mentioned in the context of s, and are thus likely to be its attributes. Further details on the construction of these sets are provided in §C. We define the *attributes rate* for a subject s in a set of tokens $\mathcal{T}$ as the portion of tokens in $\mathcal{T}$ that appear in $\mathcal{A}_s$.

Experiment We measure the attributes rate in the top $k = 50$ tokens by the *subject representation*, that is, the representation at the last-subject position, in each layer. We focus on this position as it is the only subject position that attends to all the subject positions, and thus it is likely to be the most critical (we validate this empirically in §A.3). We compare with the rate at other positions: the first subject position, the position after the subject, and the last input position (i.e., the prediction position).

Results Fig. 3 shows the results for GPT-2 and GPT-J. In both models, the attributes rate at the last-subject position is increasing throughout the layers, and is substantially higher than at other positions in the intermediate-upper layers, reaching close to 50%. This suggests that, during inference, the model constructs attribute-rich subject representations at the last subject-position. In addition, critical information from these representations propagates to the prediction, as this range of layers corresponds to the peak of critical information observed by blocking the attention edges to the prediction (§5).

We have seen that the representation of the subject encodes many terms related to it. A natural question that arises is where these terms are extracted from to enrich that representation. In principle, there are three potential sources (Mickus et al., 2022), which we turn to analyze in the next sections: the static embeddings of the subject tokens (§6.2) and the parameters of the MHSA and MLP sublayers (§6.3).

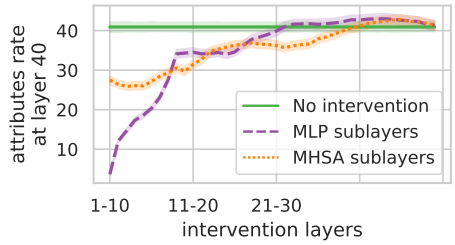

Figure 4: The attributes rate of the subject representation with and without canceling updates from the MLP and MHSA sublayers in GPT-2.

6.2 Attribute Rate in Token Embeddings

We test whether attributes are already encoded in the static embeddings of the subject tokens, by measuring the attributes rate, as in §6.1. Concretely, let $t_1, ..., t_{|s|}$ be the tokens representing a subject s (e.g. `Piet, ro, Men, nea` for "Pietro Mennea"), and denote by $\bar{\mathbf{e}} := \frac{1}{|s|}\sum_{i=1}^{|s|} \mathbf{e}_{t_i}$ their mean embedding vector, where $\mathbf{e}_{t_i}$ is the embedding of t_i. We compute the attributes rate in the top $k = 50$ tokens by each of $\mathbf{e}_{t_i}$ and by $\bar{\mathbf{e}}$. We find that the highest attributes rate across the subject's token embeddings is 19.3 on average for GPT-2 and 28.6 in GPT-J, and the average rate by the mean subject embedding is 4.1 in GPT-2 and 11.5 in GPT-J. These rates are considerably lower than the rates by the subject representations at higher layers (Fig. 3). *This suggests that while static subject-token embeddings encode some factual associations, other model components are needed for extraction of subject-related attributes.*

6.3 Subject Representation Enrichment

We next assess how different sublayers contribute to the construction of subject representations through causal interventions.

Method: Sublayer Knockout To understand which part of the transformer layer "adds" the information about attributes to the representation, we simply zero-out the two key additive elements: the MHSA and MLP sublayers. Concretely, we zero-out updates to the last subject position from each MHSA and MLP sublayer, for 10 consecutive layers. Formally, when intervening on the MHSA (MLP) sublayer at layer ℓ, we set $\mathbf{a}_i^{\ell'} = \mathbf{0}$ ($\mathbf{m}_i^{\ell'} = \mathbf{0}$) for $\ell' = \ell, ..., \min\{\ell+9, L\}$ (see Eq. 1). For each intervention, we measure the effect on the attributes rate in the subject representation at some specific layer $\bar{\ell}$, where attribute rate is high.

Results Results with respect to layer $\bar{\ell} = 40$ in GPT-2 are shown in Fig. 4, showing that canceling the early MLP sublayers has a destructive effect on the subject representation's attributes rate, decreasing it by ∼88% on average. In contrast, canceling the MHSA sublayers has a much smaller effect of <30% decrease in the attributes rate, suggesting that the MLP sublayers play a major role in creating subject representations. Results with respect to other layers and for GPT-J show similar trends and are provided in §C and §E, respectively. We further analyze this by inspecting the MLP updates, showing they promote subject-related concepts (§F).

Notably, these findings are consistent with the view of MLP sublayers as key-value memories (Geva et al., 2021; Dai et al., 2022) and extend recent observations (Meng et al., 2022a; Wallat et al., 2020) that factual associations are stored in intermediate layers, showing that they are spread across the early MLP sublayers as well.

7 Attribute Extraction via Attention

The previous section showed that the subject representation is enriched with information throughout the early-middle layers. But recall that in our prediction task, only one specific attribute is sought. How is this attribute extracted and at which point? We next show that (a) attribute extraction is typically carried out by the MHSA sublayers (§7.1) when the last position attends to the subject, (b) the extraction is non-trivial as it reduces the attribute's rank by the subject representation considerably (§7.2) and it depends on the subject enrichment process (§7.3), and (c) the relevant subject-attribute mappings are often stored in the MHSA parameters (§7.4). This is in contrast to commonly held belief that MLPs hold such information.

7.1 Attribute Extraction to the Last Position

Recall that the critical information from the subject propagates when its representation encodes many terms related to it, and after the critical information flow from the relation positions (§5). Thus, the last-position representation at this point can be viewed as a relation query to the subject representation. We therefore hypothesize that the critical information that flows at this point is the attribute itself.

Experiment: Extraction Rate To test this hypothesis, we inspect the MHSA updates to the last position in the vocabulary, and check whether the